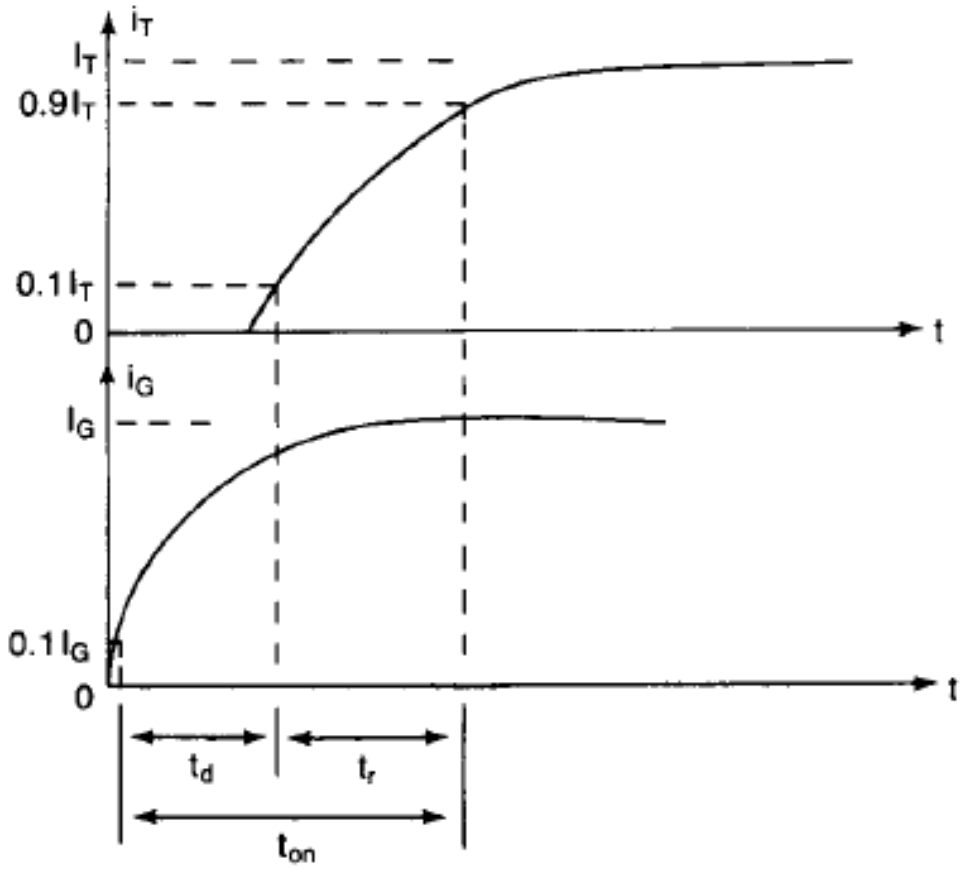
Lecture 8

Dynamic Turn ON

1. The gate signal can be removed after the thyristor is turned on. A continuous gating signal would increase the power loss in the gate junction.

2. While the thyristor is reversed biased, there should be no gate signal; otherwise, the thyristor may fall due to increase leakage current.

3. The width of gate pulse, t_G, must be longer than the time required for the anode current to rise to the holding current value. In practice, the pulse width, t_G, is normally made more than the turn-on time of the thyristor, t_{on}.



Dynamic Turn ON

Example 14-1

The holding current of thyristors in the single-phase full converter of Fig. 14-3a is I_H = 500 mA and the delay time is t_d = 1.5 μs. The converter is supplied from a 120-V 60-Hz supply and has a load of L = 10 mH and R = 10 Ω. The converter is operated with a delay angle of α = 30°. Determine the minimum value of gate pulse width, t_G.

Dynamic Turn ON

Solution $I_H = 500\ \text{mA} = 0.5\ \text{A}$, $t_d = 1.5\ \mu\text{s}$, $\alpha = 30^\circ = \pi/6$, $L = 10\ \text{mH}$, and $R = 10\ \Omega$. The instantaneous value of the input voltage is $v(t) = V_m \sin \omega t$, where $V_m = \sqrt{2} \times 120 = 169.7\ \text{V}$.

At $\omega t = \alpha$,

$$V_1 = v(\omega t = \alpha) = 169.7 \times \sin \frac{\pi}{6} = 84.85\ \text{V}$$

The rate of rise of anode current, di/dt, at the instant of triggering is approximately,

$$\frac{di}{dt} = \frac{V_1}{L} = \frac{84.85}{10 \times 10^{-3}} = 8485\ \text{A/s}$$

If di/dt is assumed constant for a short time after the gate triggering, the time required for the anode current to rise to the level of holding current, t_1, is calculated from $t_1 \times (di/dt) = I_H$ or $t_1 \times 8485 = 0.5$ and this gives $t_1 = 0.5/8485 = 58.93\ \mu\text{s}$. Therefore, the minimum width of gate pulse is

$$t_G = t_1 + t_d = 58.93 + 1.5 = 60.43\ \mu\text{s}$$

SCR Turn OFF Methods

Introduction

- SCR turned ON by applying appropriate positive gate voltage between the gate and cathode, but it cannot be turned OFF through the gate terminal
- SCR brought back to the forward blocking state by reducing the anode current below the holding current level.
- The turn OFF process of an SCR is called **commutation**.
- Commutation means the transfer of currents from one path to another. So the commutation circuit reducing the forward current to zero so as to turn OFF the SCR or Thyristor.

To turn OFF the conducting SCR the below conditions must be satisfied.

- The anode current of SCR must be reduced to zero or below the level of holding current and then,
- A sufficient reverse voltage must be applied across the SCR to regain its forward blocking state

SCR Turn OFF Methods

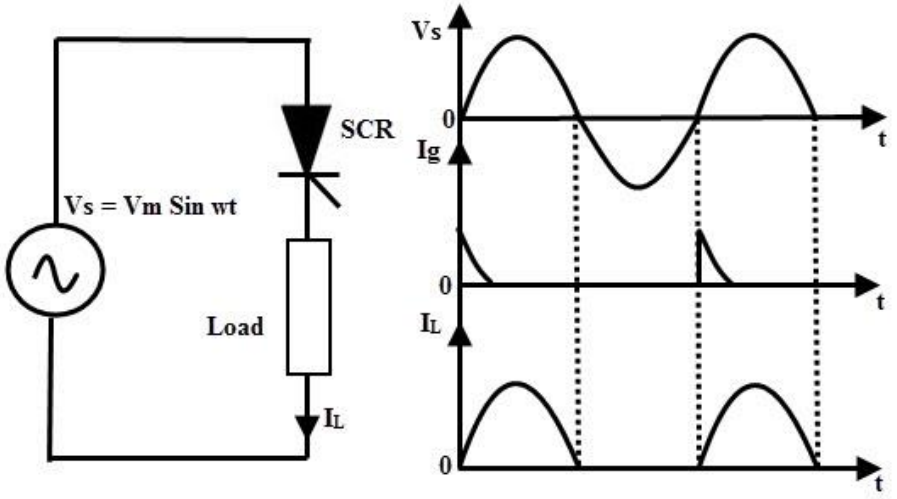
The reverse voltage which causes to commutate the SCR is called commutation voltage. Depending on the commutation voltage located, the commutation methods are classified into two major types.

1) Forced commutation

2) Natural commutation

Natural Commutation

- The source of commutation voltage is the supply source itself
- On AC supply, at every end of the positive half cycle the anode current goes through the natural current zero and also immediately a reverse voltage is applied across the SCR. These are the conditions to turn OFF the SCR.
- This method is also called as source, or line, or class F commutation
- This commutation is possible with line commutated inverters, controlled rectifiers, cyclo converters and AC voltage regulators because the supply is the AC source in all these converters.

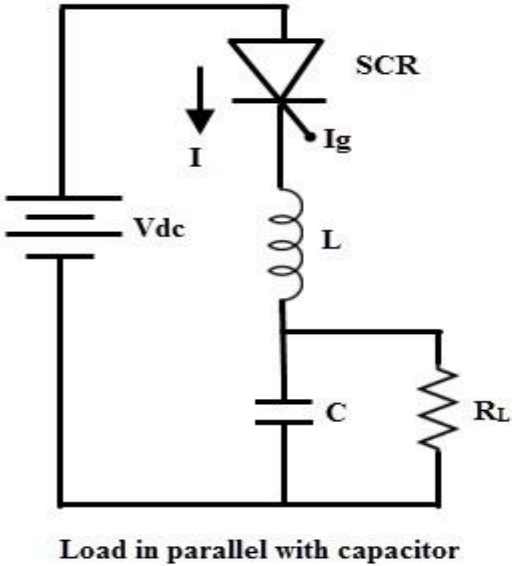


Forced Commutation

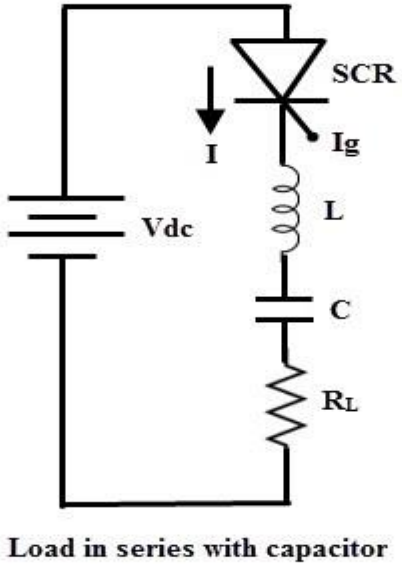
- In case of DC circuits, there is no natural current zero to turn OFF the SCR. In such circuits, forward current must be forced to zero with an external circuit to commutate
- This circuit consist of components like inductors and capacitors called as commutating components.
- These components cause reverse voltage across the SCR that immediately bring the current in the SCR to zero
- This method further classified, based on how zero current achieved and arrangement of the commutating components,

Class A Commutation

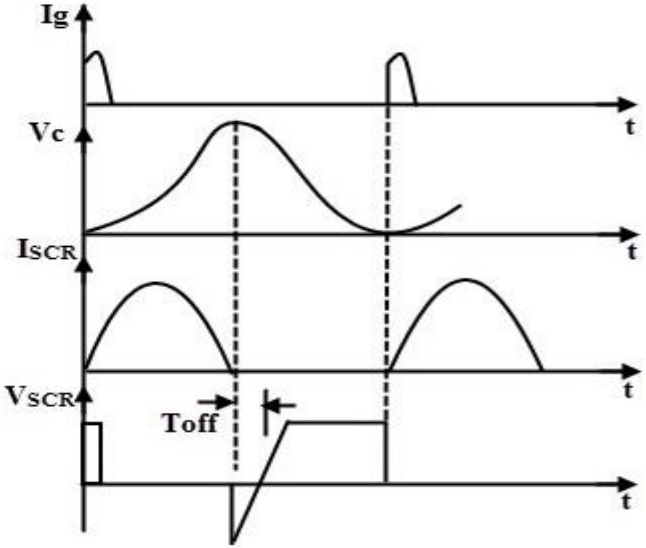
- ➔ It is also known as self commutation, or resonant commutation, or load commutation.
- ➔ The source of commutation voltage is in load which is an underdamped R-L-C ckt with a DC supply
- ➔ The value of load resistance and commutating components are so selected that they forms a under damped resonant circuit to produce natural zero
- ➔ Once capacitor is fully charged the SCR becomes reverse biased and hence the commutation of the device.
- ➔ It's simple and reliable method for high freq operation.above 1000 Hz and high value of L & C



Load in parallel with capacitor

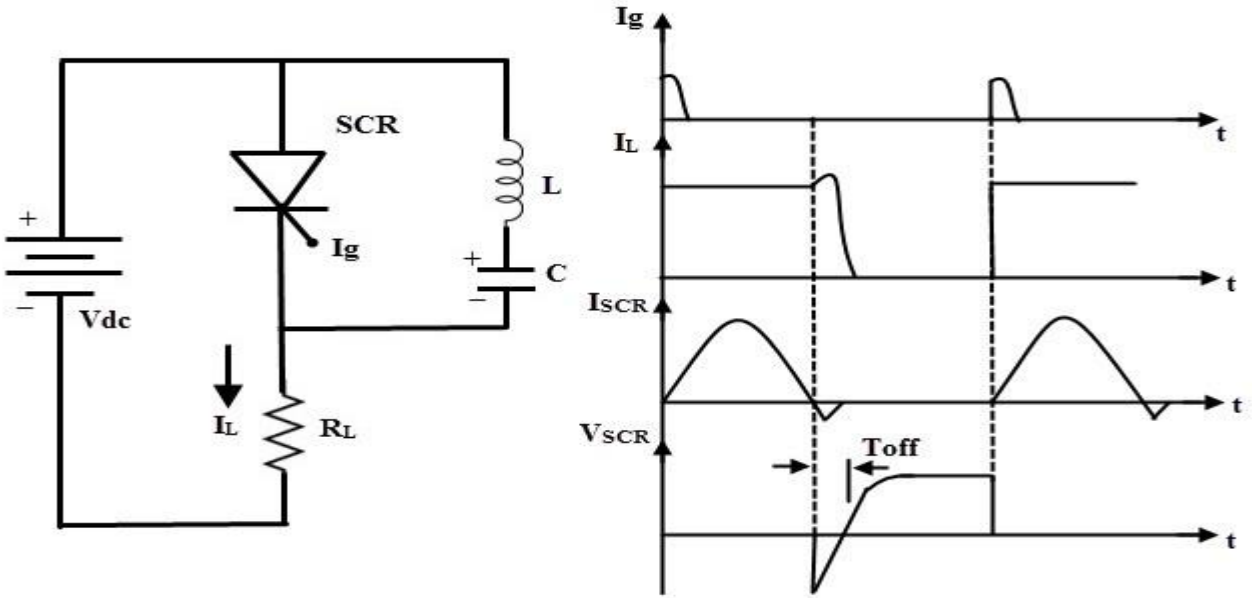


Load in series with capacitor



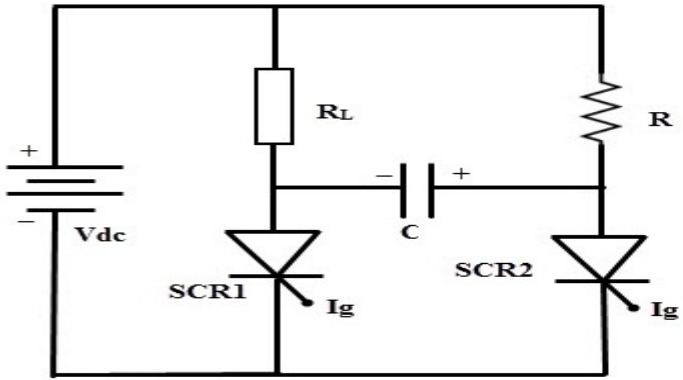
Class B Commutation

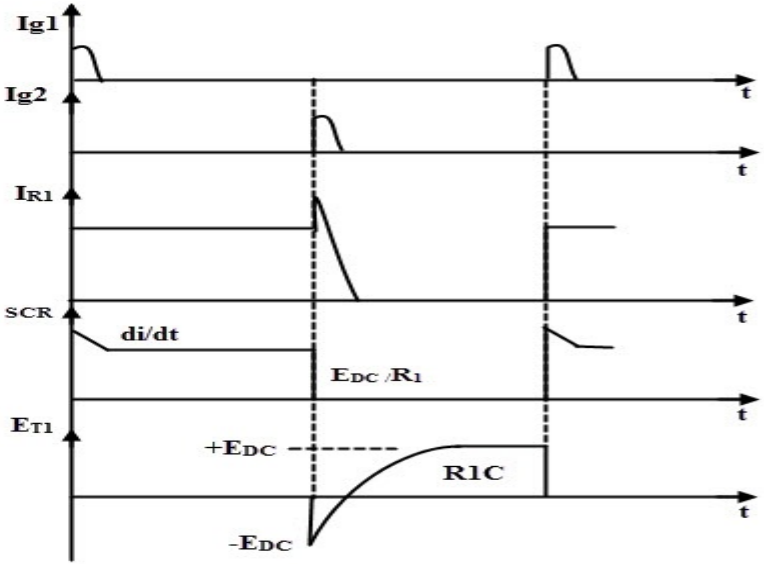
- ➔ It is also a self commutation circuit, commutation is achieved automatically by L and C
- ➔ In this, LC resonant circuit is connected across the SCR but not in series with load, hence the L and C components do not carry the load current
- ➔ In this process, SCR is turned ON for some time and then automatically turned OFF for some time.
- ➔ This is continuous process and desired freq depends on the values of L& C.It mostly used in chopper circuits.



Class C Commutation

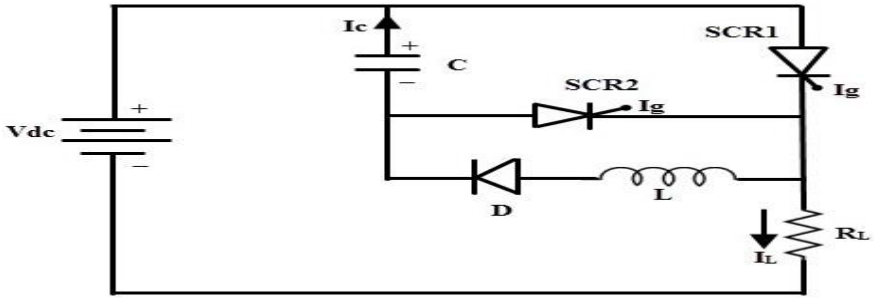
- In this method, the main SCR is connected in series with load and an additional SCR is connected in parallel with main SCR. This method also called complementary commutation
- SCR turns OFF with a reverse voltage of a charged capacitor.
- This process is mainly used in single phase inverters with a centre tapped transformers.
- This is a very reliable method of commutation and it is also useful even at frequencies below 1000Hz.

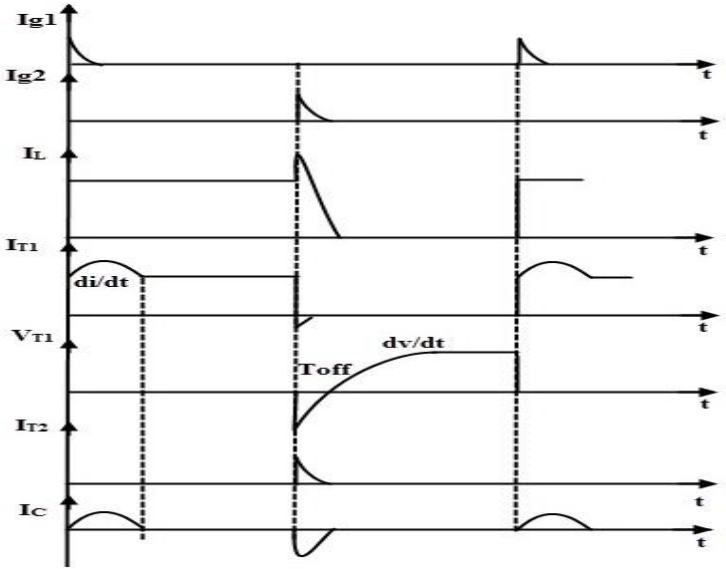




Class D Commutation

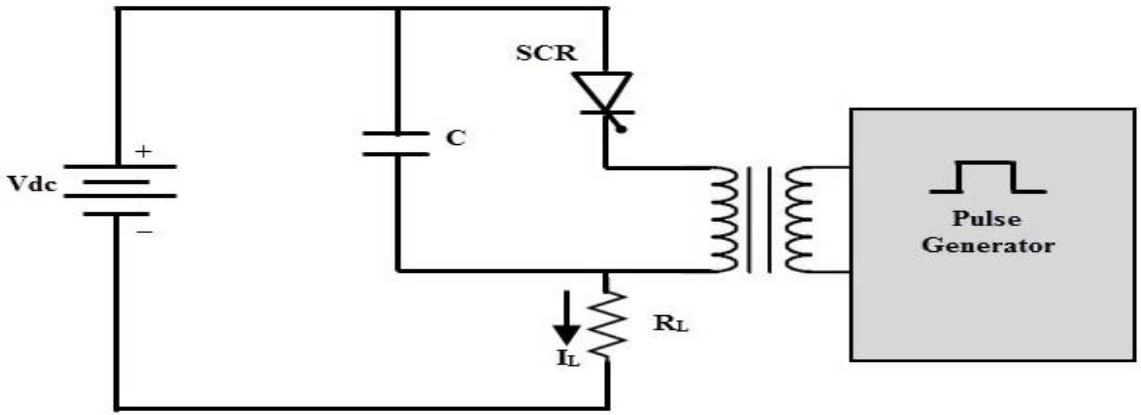
- ➔ It is also called auxiliary commutation because it uses an auxiliary SCR to switch charged capacitor.
- ➔ The main SCR with load resistance forms the power circuit while the diode D, inductor L and SCR2 forms the commutation circuit.
- ➔ SCR2 must be triggered first.in order to charge capacitor and turned OFF when fully charged
- ➔ SCR1 is triggered, the current flows in two directions; one is the load current path and another one is commutation current path.As capacitor completely discharges, its polarities reversed
- ➔ Due to the presence of diode the reverse discharge is not possible

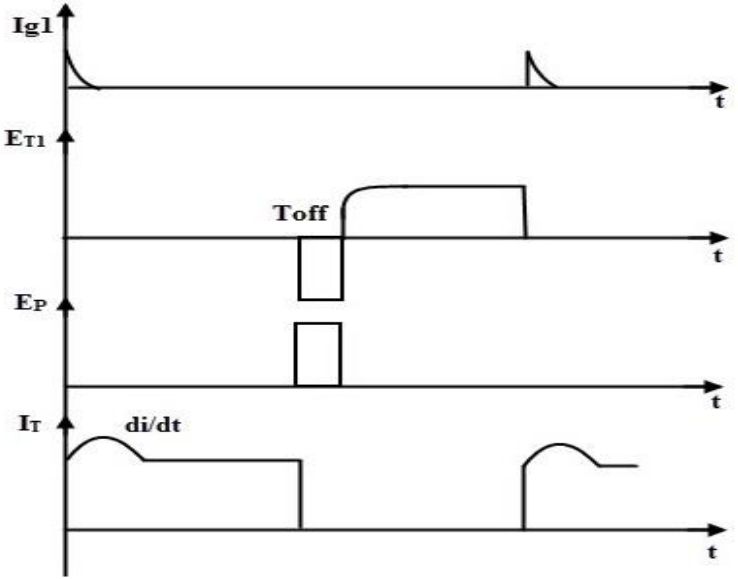




Class E Commutation

- ➔ It is also known as external pulse commutation
- ➔ It uses pulse transformer to produce commutating pulse and is designed with tight coupling
- ➔ To commutated, pulse duration should be equal to the turn OFF time of SCR
- ➔ SCR is triggered, load current flows through the pulse transformer. If the pulse is applied to the primary of the pulse transformer, emf is induced in the secondary of the pulse transformer.
- ➔ This induced voltage is applied across the SCR in reverse polarity and hence the SCR turned OFF.





Dynamic Turn OFF Switching Characteristics

- The transition of an SCR from forward conduction state to forward blocking state is called as turn OFF or commutation of SCR.
- Once the SCR starts conducting, the gate has no control over it to bring back to forward blocking or OFF state.
- To turn OFF the SCR, the current must be reduced to a level below the holding current of SCR.
- But if we apply the forward voltage immediately after the current zero of SCR, it starts conducting again even without gate triggering.
- This is due to the presence of charge carriers in the four layers.
- It is necessary to apply the reverse voltage, over a finite time across the SCR to remove the charge carriers.

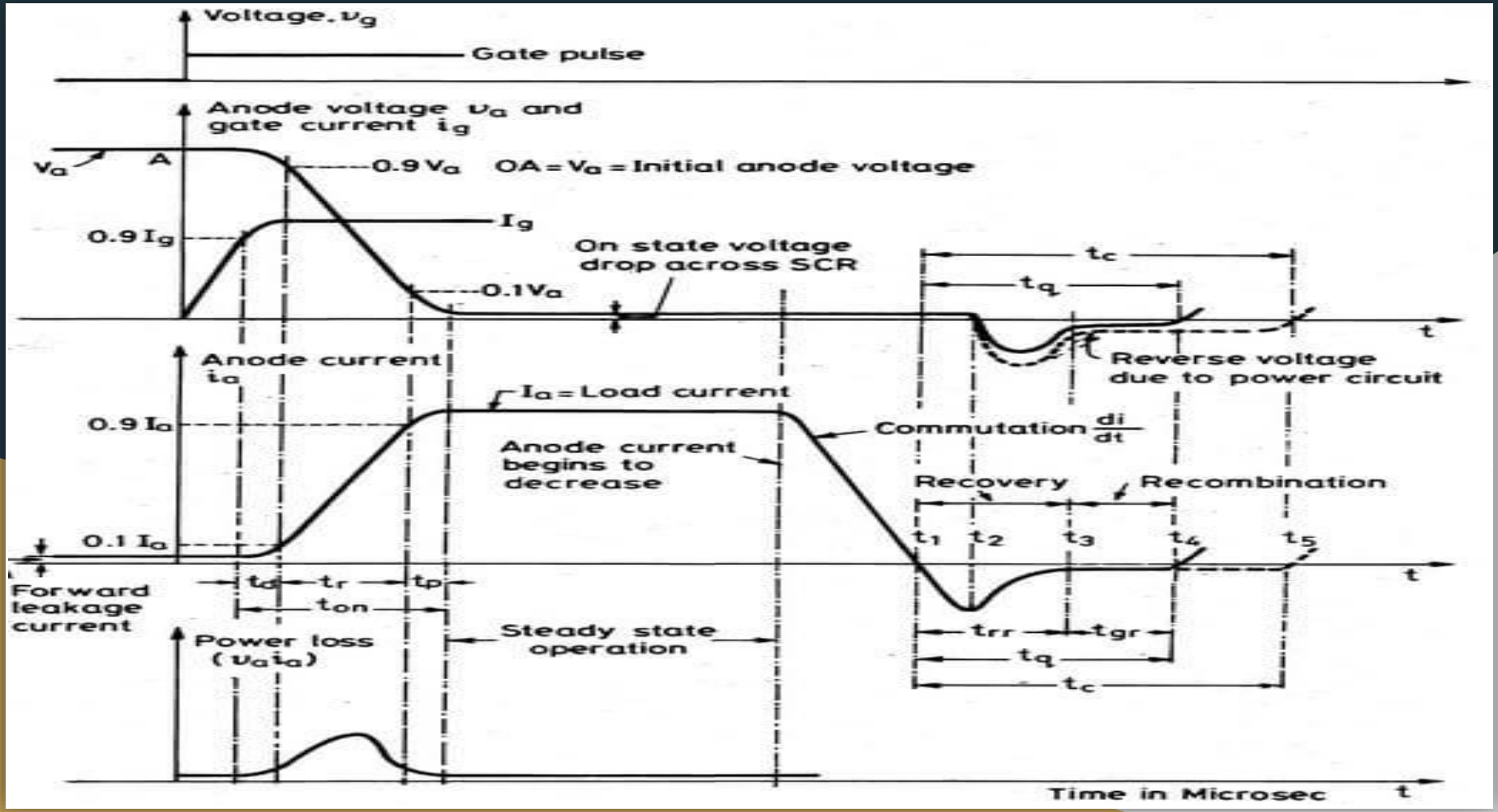
Turn OFF

- The turn OFF time is defined as the time between the instant the anode current becomes zero and the instant at which the SCR retains the forward blocking capability.
- The excess charge carriers from the four layers must be removed to bring back the SCR to forward blocking mode.
- This process takes place in two stages.
- In a first stage excess carriers from outer layers are removed and in second stage excess carriers in the inner two layers are to be recombined.
- the total turn OFF time t_q is divided into two intervals; reverse recovery time t_{rr} and gate recovery time t_{gr}.

$$\mathbf{t_q = t_{rr} + t_{gr}}$$

Reverse Recovery & Gate Recovery Time

- The time t_1 to t_3 is called as reverse recovery time.
- At the instant t_1 the anode current is zero and builds up in the reverse direction which is called as reverse recovery current
- This current removes the excess charge carriers from outer layers during the time t_1 to t_3.
- At instant t_3, junctions J_1 and J_3 are able to block the reverse voltage.
- Still the SCR is not yet able to block the forward voltage due to the presence of excess charge carriers in junction J_2
- These carriers can be disappeared only by the way of recombination and this could be achieved by maintaining a reverse voltage across the SCR.
- During the time t_3 to t_4, the recombination of charges takes place and at the instant t_4, junction J_2 completely recovers. This time is called gate recovery time t_{gr}.


Voltage, v_g
Gate pulse
Anode voltage v_a and gate current i_g
V_a
A
0.9 V_a
OA = V_a = Initial anode voltage
I_g
0.9 I_g
On state voltage drop across SCR
0.1 V_a
t_c
t_q
t
Reverse voltage due to power circuit
Anode current i_a
I_a = Load current
0.9 I_a
Commutation $\frac{di}{dt}$
Anode current begins to decrease
Recovery
Recombination
0.1 I_a
t_1 t_2 t_3 t_4 t_5
Forward leakage current
t_d t_r t_p
t_{on}
Steady state operation
Power loss ($v_a i_a$)
t_{rr} t_{gr}
t_q
t_c
Time in Microsec
t

Important Points

- From the figure the turn OFF time is the time interval between the t_4 and t_1. Generally, this time varies from 10 to 100 microseconds.
- The time required by the commutation circuit to apply the reverse voltage to commutate the SCR is called the circuit turn OFF time (t_c).
- For a safety margin or reliable commutation, this t_c must be greater than the t_q otherwise commutation failure occurs.
- The SCRs which have slow turn OFF time (50 to 100 microseconds) are called as converter grade SCRs. These are used in phase controlled rectifiers, cyclo converters, AC voltage regulators, etc.
- The SCRs which have fast turn OFF time (3 to 50 microseconds) are inverter grade SCRs. These are costlier compared to converter grade and are used in choppers, force commutated converters and inverters